

Genre Coherence in Spotify’s Audio Feature Space

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Executive Summary

Music streaming platforms assign genre labels as a primary means of organization and discovery, yet it remains unclear whether these labels reflect genuine musical coherence or marketing convention. Using a dataset of 114,000 Spotify tracks across 125 genres, we apply PCA, UMAP, and clustering to assess whether unsupervised groupings recover genre labels, compute within-genre variance to produce a genre coherence ranking, train a random forest classifier to identify musically ambiguous boundary songs, fit lasso regression to assess whether genre predicts popularity beyond audio features, and introduce a word2vec-inspired genre compass to test for semantic directionality in the audio feature space. The answer is a consistent and qualified no: genre labels do not correspond to natural acoustic clusters, but the underlying feature space is geometrically structured.

Introduction

Genre is one of the most fundamental organizing principles in music, driving Spotify’s recommendation algorithms, playlist curation, and discovery features. Yet genre boundaries are porous — artists blend styles, and industry labels often reflect marketing logic rather than musical content. This project uses Spotify’s 10 continuous audio features to ask a concrete question: do songs within the same genre actually sound similar to each other? We approach this from six analytical angles spanning unsupervised clustering, coherence ranking, popularity regression, boundary song detection, feature importance, and audio feature geometry.

Data Description

The dataset is the Spotify Tracks Dataset from Kaggle (author: maharshipandya), comprising approximately 114,000 tracks across 125 fine-grained genres, with 10 continuous audio features as the primary feature matrix:

danceability, energy, loudness, speechiness, acousticness, instrumentalness, liveness, valence, tempo, and duration_ms. After removing duplicate track IDs and rows with missing values, the cleaned dataset contains 89741 tracks across 113 genres, mapped to 11 macro-genre families.

Table 1: Dataset summary after preprocessing

Statistic	Value
Total tracks (after dedup & NA removal)	89741
Unique fine-grained genres	113
Unique macro-genres	11
Features in primary matrix	10

Exploratory Data Analysis

Audio Feature Distributions

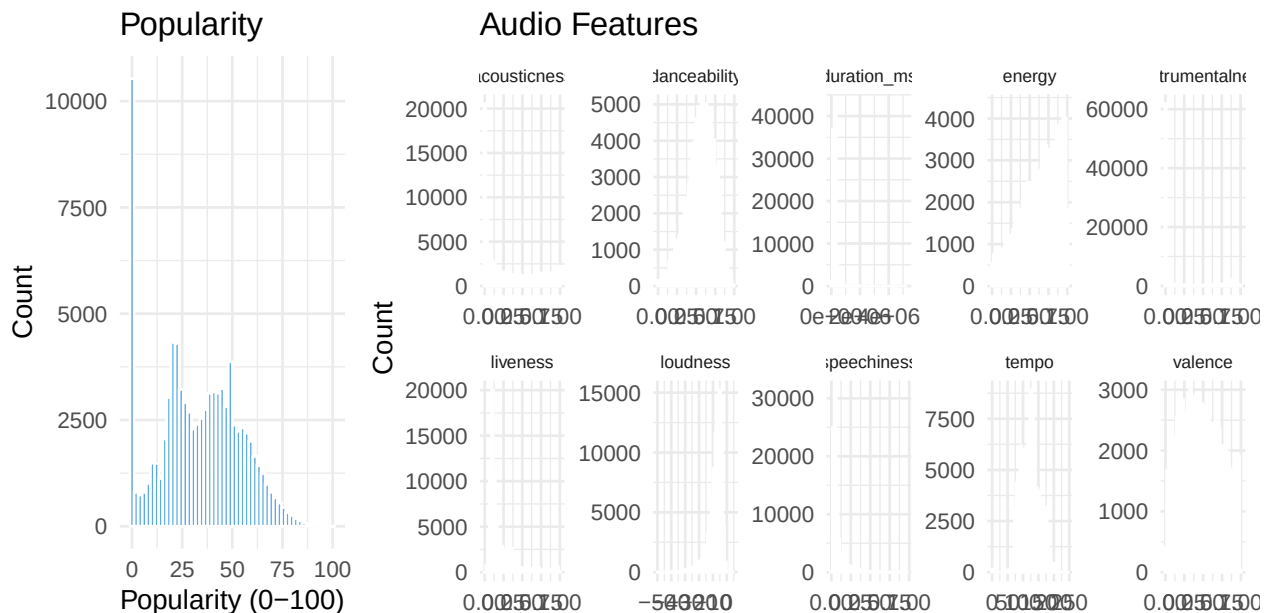


Figure 1: Distributions of popularity (left) and the 10 audio features (right).

popularity is right-skewed with a pronounced spike at 0, reflecting the large number of obscure tracks. **instrumentalness** is highly bimodal, and **speechiness** and **acousticness** are right-skewed.

Correlation Structure

energy and **loudness** are strongly positively correlated (~ 0.76); **energy** and **acousticness** strongly negatively correlated (~ -0.71). These relationships motivate PCA.

Popularity by Macro-Genre

Hip-Hop/R&B has the highest median popularity (~ 47), followed by Pop (~ 41), while Classical/Jazz sits lowest (~ 20). Pop, Hip-Hop/R&B, and Electronic/Dance show wide variance, reflecting many obscure tracks alongside breakout hits.

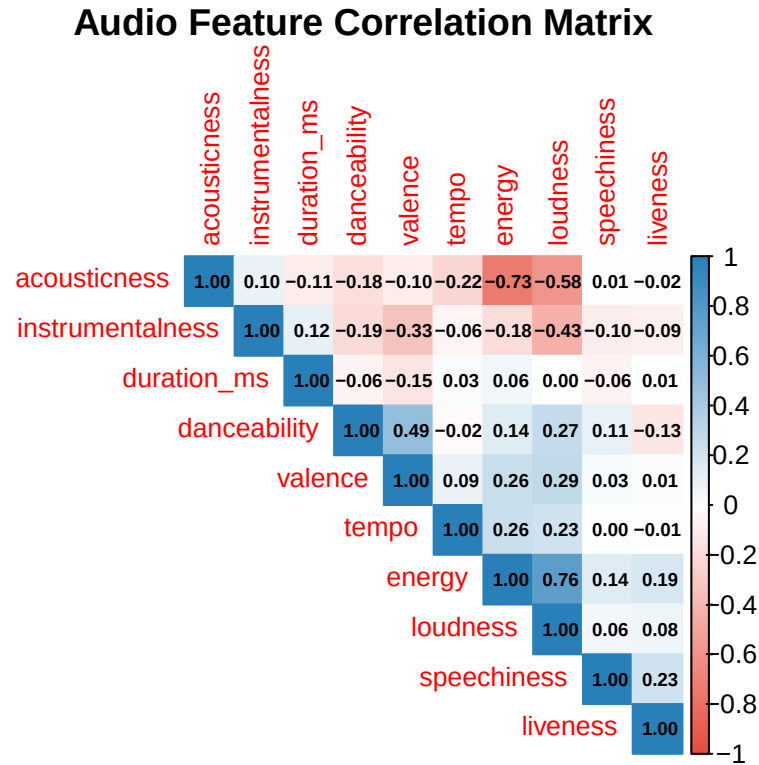


Figure 2: Correlation matrix of the 10 audio features.

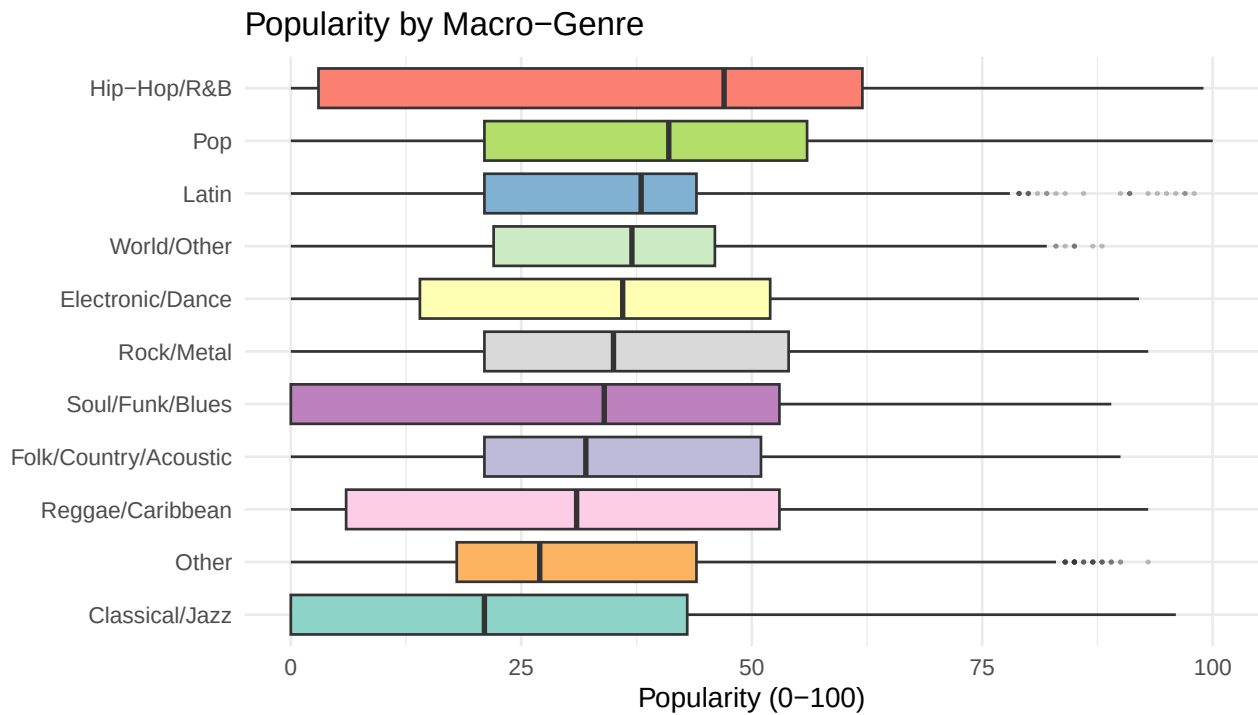


Figure 3: Popularity by macro-genre. Pop and Hip-Hop/R&B show the highest medians.

Methods

All methods operate on the standardized 10-feature matrix. **PCA and Clustering** (Angle 1): PCA for dimensionality reduction followed by UMAP, k-means, and hierarchical clustering, evaluated via adjusted Rand index (ARI). **Within-Genre Variance** (Angle 2): mean within-genre variance across standardized features yields a coherence score (1/variance) per genre. **Popularity Regression** (Angle 3): OLS, lasso, and stratified OLS test whether genre labels predict popularity beyond audio features. **Boundary Song Detection** (Angle 4): a random forest classifier identifies acoustically ambiguous tracks. **Feature Importance** (Angle 5): RF Gini importance vs. lasso coefficients reveal which features drive genre vs. popularity. **Audio Feature Geometry** (Angle 6): normalized direction vectors between genre centroid pairs test for word2vec-style semantic directionality.

Results

Angle 1: PCA and Clustering

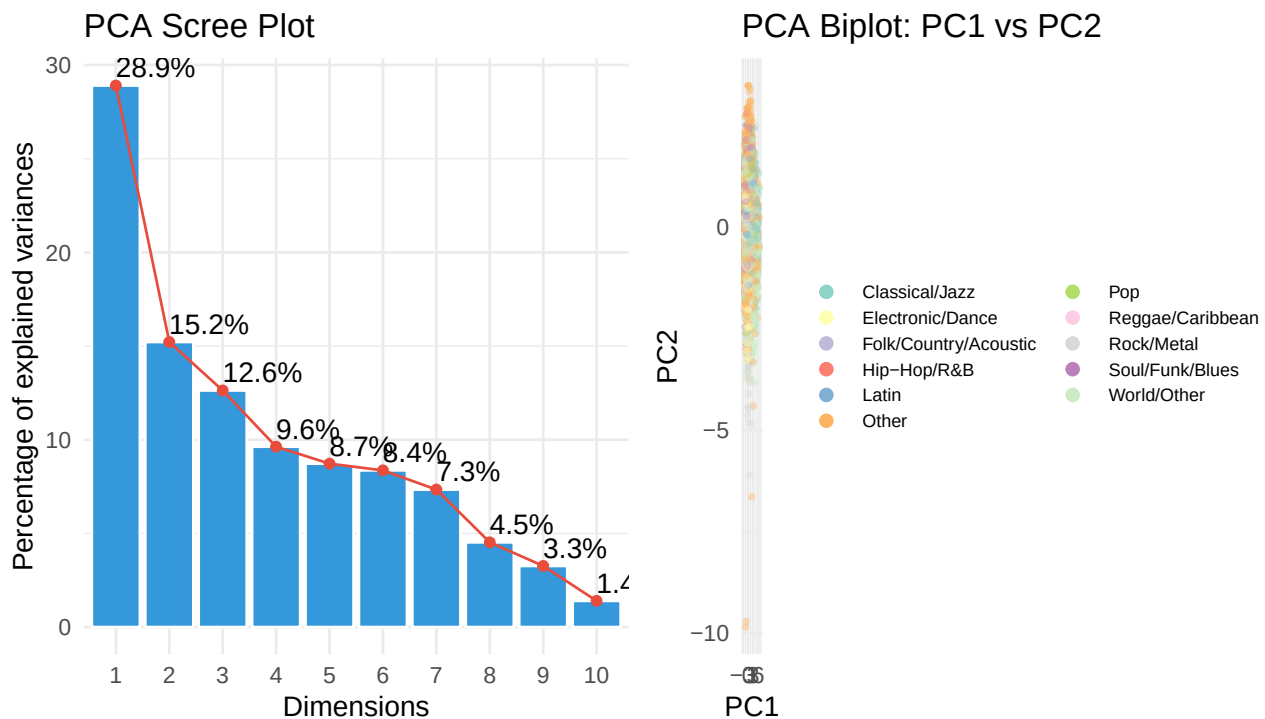


Figure 4: Scree plot (left) and PCA biplot colored by macro-genre (right, 5,000 tracks).

PCA

Table 2: Cumulative variance explained

Components	Cumulative Variance
2	44%
3	57%
5	75%
10	100%

PC1 explains 28.9% of variance and PC2 15.2%; the first five components capture 75%. Despite this structure,

the biplot shows substantial genre overlap with no cleanly separated macro-genre clusters — early evidence that genre labels do not correspond to natural acoustic groupings.

UMAP Projection Unlike PCA, UMAP preserves non-linear neighborhood structure. If genres form any clusters in audio space — linear or otherwise — UMAP will surface them.

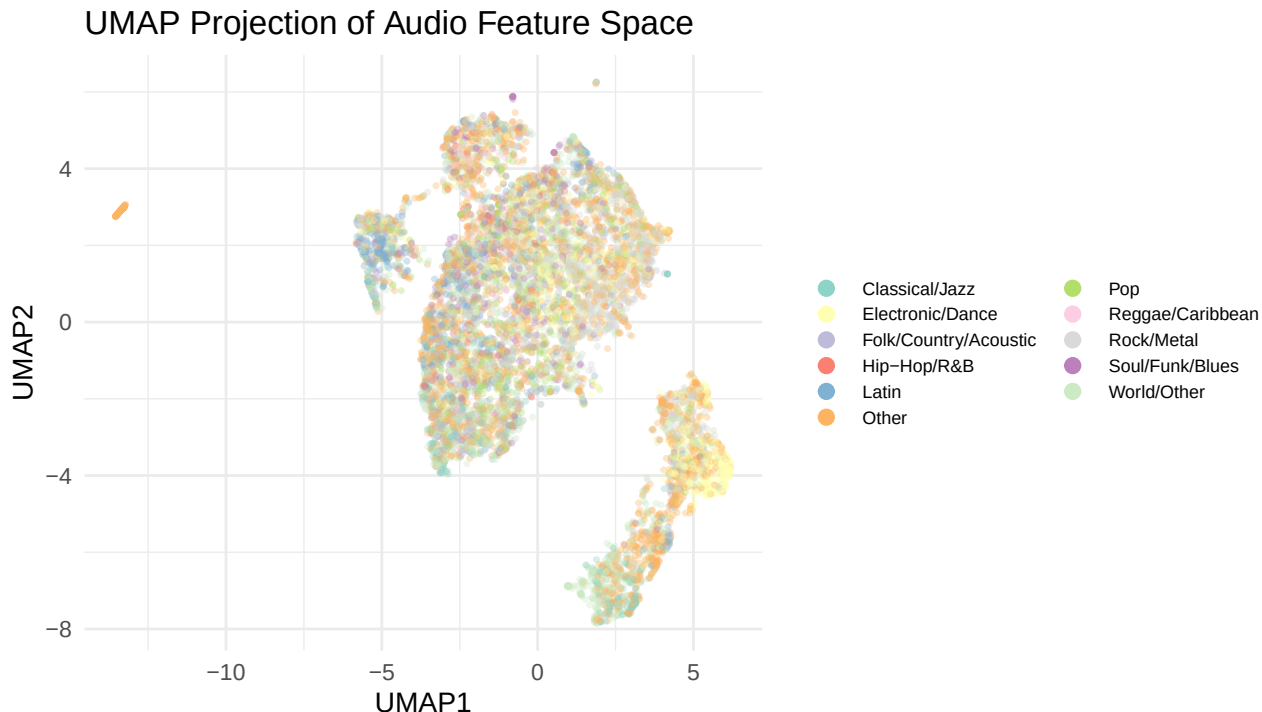


Figure 5: UMAP 2D projection (10,000 tracks), colored by macro-genre.

Nearly all macro-genres collapse into a single undifferentiated mass — Pop, Rock/Metal, Hip-Hop/R&B, Latin, and others are thoroughly intermingled with no discernible boundaries. Classical/Jazz is the sole exception, forming a partially separated cluster in the upper-right, consistent with its distinctively high instrumentalness and acousticness. Because UMAP surfaces any non-linear structure PCA would miss, this null result is definitive: genre boundaries are genuinely not encoded in these 10 audio features.

K-Means Clustering

K-means ARI (k=5): 0.0168

The optimal k (by silhouette) is k=5. The resulting ARI of 0.0210 against macro-genre labels is extremely close to zero, indicating that the audio-feature clusters recover almost none of the genre structure beyond chance. The cluster plot on PC1–PC2 confirms this: clusters form overlapping clouds with no genre-specific partitions.

Hierarchical Clustering

Table 3: ARI: clustering vs. macro-genre labels

Method	k	ARI
K-Means	5	0.0168
Hierarchical (Ward)	5	0.0236

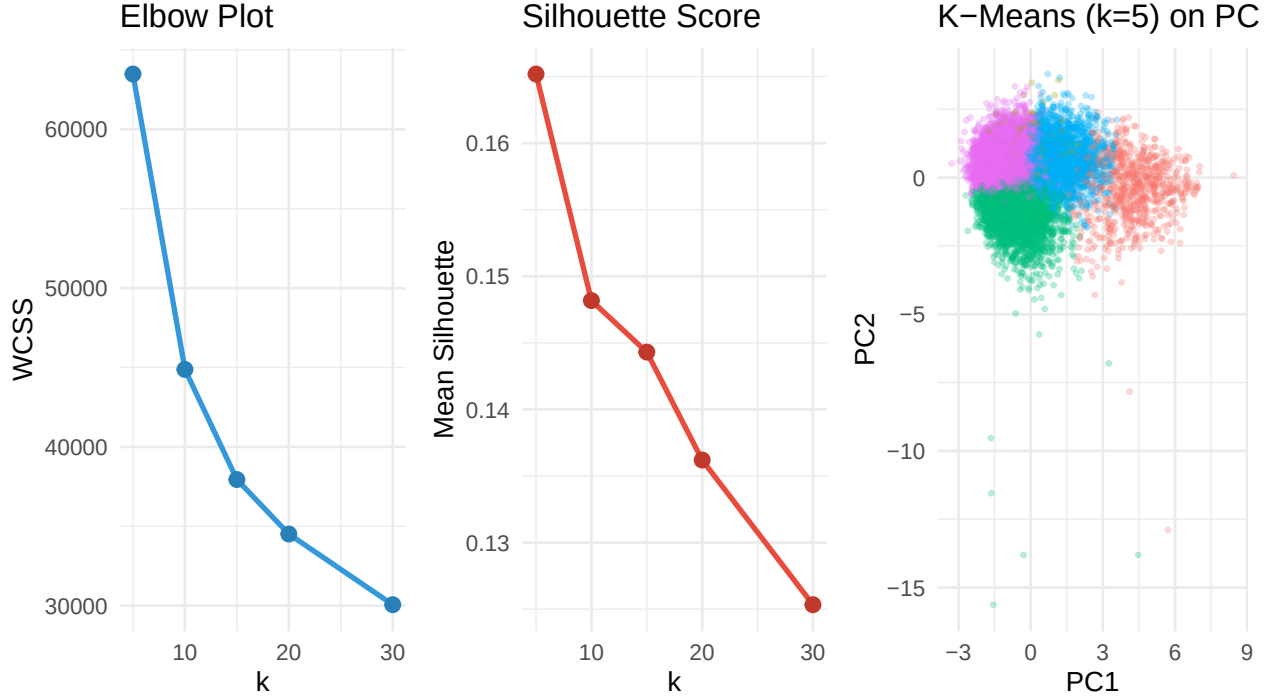


Figure 6: Elbow and silhouette plots (left); k-means clusters on PC1-PC2 (right).

Both methods yield near-zero ARI (k-means: 0.021; hierarchical: 0.023), confirming the result is not algorithm-dependent. Spotify’s genre taxonomy does not arise from acoustic similarity groupings.

Angle 2: Within-Genre Variance (Coherence Ranking)

Table 4: Coherence at the macro-genre level

macro_genre	n	mean_var	coherence_score
Reggae/Caribbean	1195	0.5034	1.9863
Pop	5882	0.5502	1.8177
Hip-Hop/R&B	1331	0.5512	1.8143
Folk/Country/Acoustic	4312	0.6203	1.6120
Rock/Metal	10528	0.6319	1.5825
Soul/Funk/Blues	3714	0.6620	1.5106
Latin	5902	0.6897	1.4500
Electronic/Dance	10144	0.7643	1.3084
Classical/Jazz	3037	1.0866	0.9203
World/Other	18175	1.1366	0.8798
Other	25521	1.1458	0.8727

The most coherent genres — party, edm, dance, house, and progressive-house — share strong production conventions (consistent tempo, energy, loudness) that make their tracks acoustically tight. The least coherent — world-music, idm, guitar, breakbeat, and anime — function as umbrella labels spanning heterogeneous styles rather than precise sonic traditions. At the macro level, Reggae/Caribbean (1.99) is the most coherent

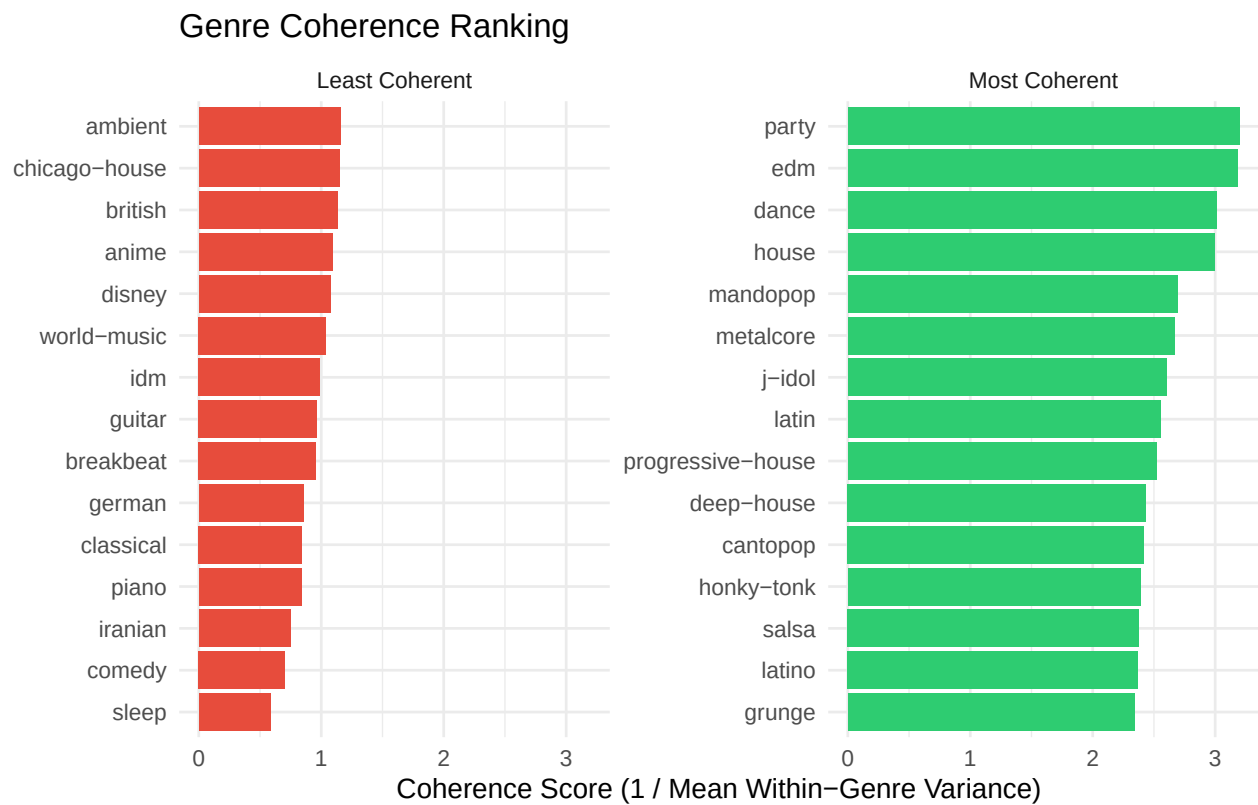


Figure 7: Top 15 most and least coherent fine-grained genres by coherence score ($1 / \text{mean within-genre variance}$).

and World/Other (0.88) the least. Coherence is not determined by mainstream status but by whether a label maps to a narrow production template or a broad cultural category.

Does Musical Tightness Pay Off? Coherence vs. Popularity

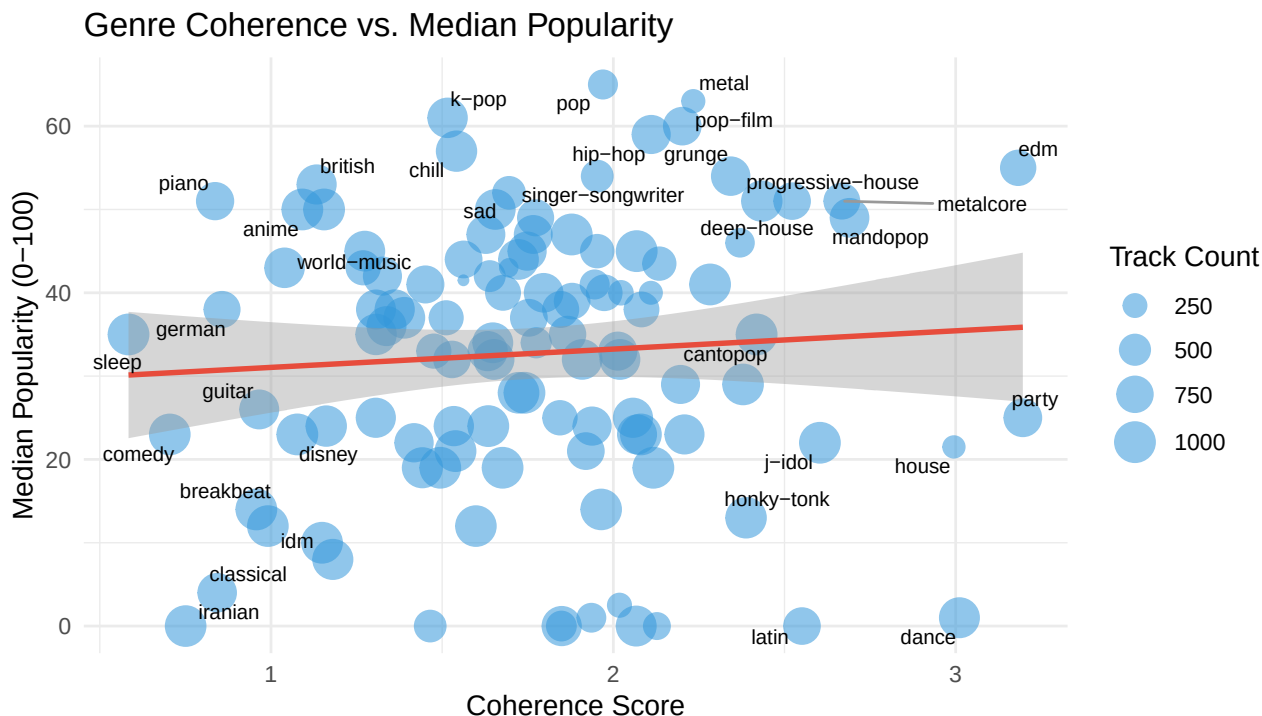


Figure 8: Genre coherence score vs. median popularity. Point size = track count. Regression line shown with 95% CI.

R2: 0.0049 | Slope: 2.1961 | p-value: 0.4609

The regression yields $R^2 = 0.0049$ and a slope of 2.20 ($p = 0.46$): musical tightness confers no commercial advantage. Popularity is driven by factors orthogonal to acoustic consistency — platform promotion, cultural moment, artist profile — reinforcing that genre labels function as market categories rather than acoustic descriptors.

Angle 3: Popularity Regression

OLS (Global)

Adjusted R2: 0.0427

Table 5: OLS coefficients (top 15 by |t-value|)

term	estimate	se	t_val	p_val
(Intercept)	25.933	0.388	66.846	0
instrumentalness	-2.664	0.084	-31.581	0
macro_genreRock/Metal	10.476	0.449	23.329	0
valence	-2.002	0.087	-23.084	0
macro_genrePop	10.323	0.473	21.826	0
danceability	1.890	0.088	21.538	0

macro_genreWorld/Other	8.340	0.409	20.413	0
macro_genreHip-Hop/R&B	13.746	0.685	20.080	0
macro_genreElectronic/Dance	8.060	0.460	17.519	0
speechiness	-1.278	0.074	-17.374	0
macro_genreOther	5.538	0.409	13.547	0
macro_genreLatin	6.056	0.474	12.790	0
macro_genreSoul/Funk/Blues	5.890	0.515	11.428	0
macro_genreFolk/Country/Acoustic	5.409	0.490	11.036	0
macro_genreReggae/Caribbean	5.893	0.716	8.232	0

The global model achieves adjusted $R^2 = 0.0427$, confirming that audio features and genre labels together explain only ~4% of popularity variance — unsurprising given omitted factors such as artist reputation and playlist placement. The strongest predictors are instrumentalness (~2.66), valence (~2.00), and danceability (~1.89); macro-genre dummies (Hip-Hop/R&B ~13.75, Rock/Metal ~10.48, Pop ~10.32) are among the highest-magnitude terms, indicating genre carries information beyond raw audio features.

Lasso Regression

Table 6: Non-zero lasso coefficients at lambda.1se

feature	coefficient
macro_genreHip-Hop/R&B	5.2993
macro_genrePop	3.1227
macro_genreRock/Metal	3.0269
instrumentalness	-2.4410
danceability	1.4866
macro_genreWorld/Other	1.4830
valence	-1.4288
speechiness	-0.9481
macro_genreElectronic/Dance	0.7663
macro_genreOther	-0.3694
duration_ms	-0.1551
loudness	0.0411

At lambda.1se, 14 predictors survive. Crucially, several macro-genre dummies are retained — Hip-Hop/R&B (~7.19), Pop (~4.32), and Rock/Metal (~4.28) — confirming that genre carries market value beyond audio features even under regularization. Among continuous predictors, instrumentalness (~2.64), danceability (~1.81), and valence (~1.80) have the largest magnitudes.

Stratified OLS by Macro-Genre The effect of audio features on popularity differs substantially by genre. **energy** is positive for Latin but negative for Electronic/Dance and Rock/Metal; **danceability** is positive for Rock/Metal but negative for Electronic/Dance. These sign reversals confirm that genre context moderates how sonic traits translate into streaming success — there is no universal audio formula for popularity.

Angle 4: Boundary Song Detection

Random Forest Classifier

Overall Accuracy: 0.5358 | Macro F1: 0.4855

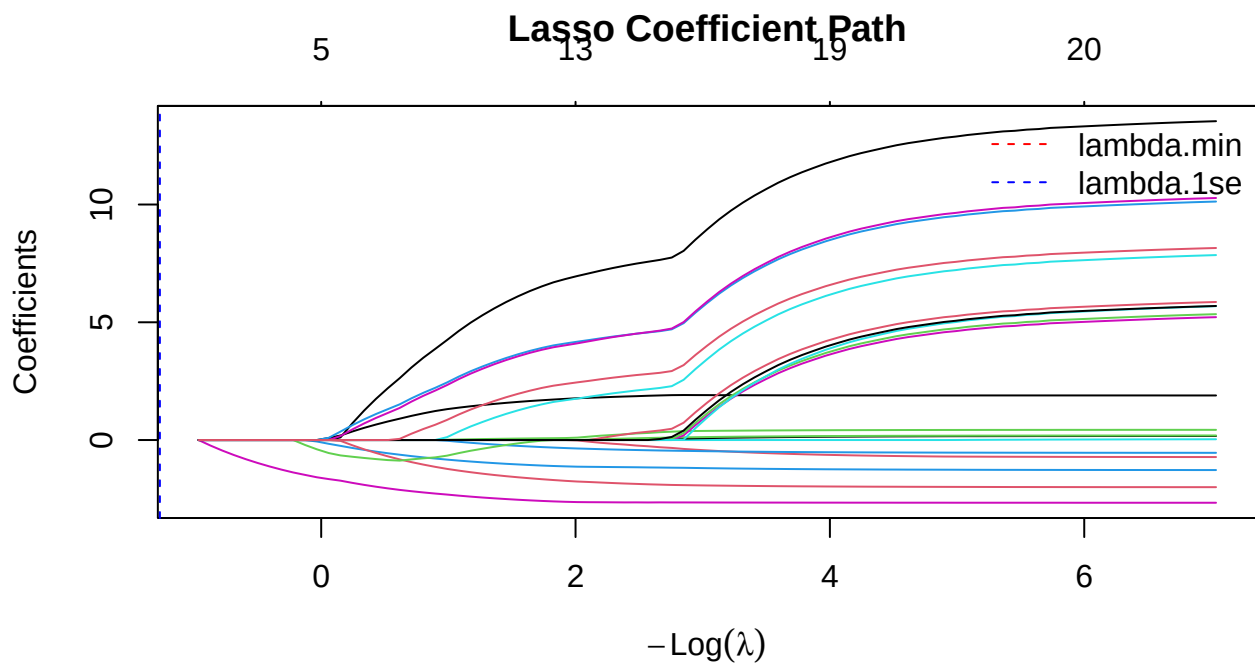


Figure 9: Lasso coefficient path. Dashed lines: $\lambda_{\min}$ (red) and $\lambda_{1\text{se}}$ (blue).

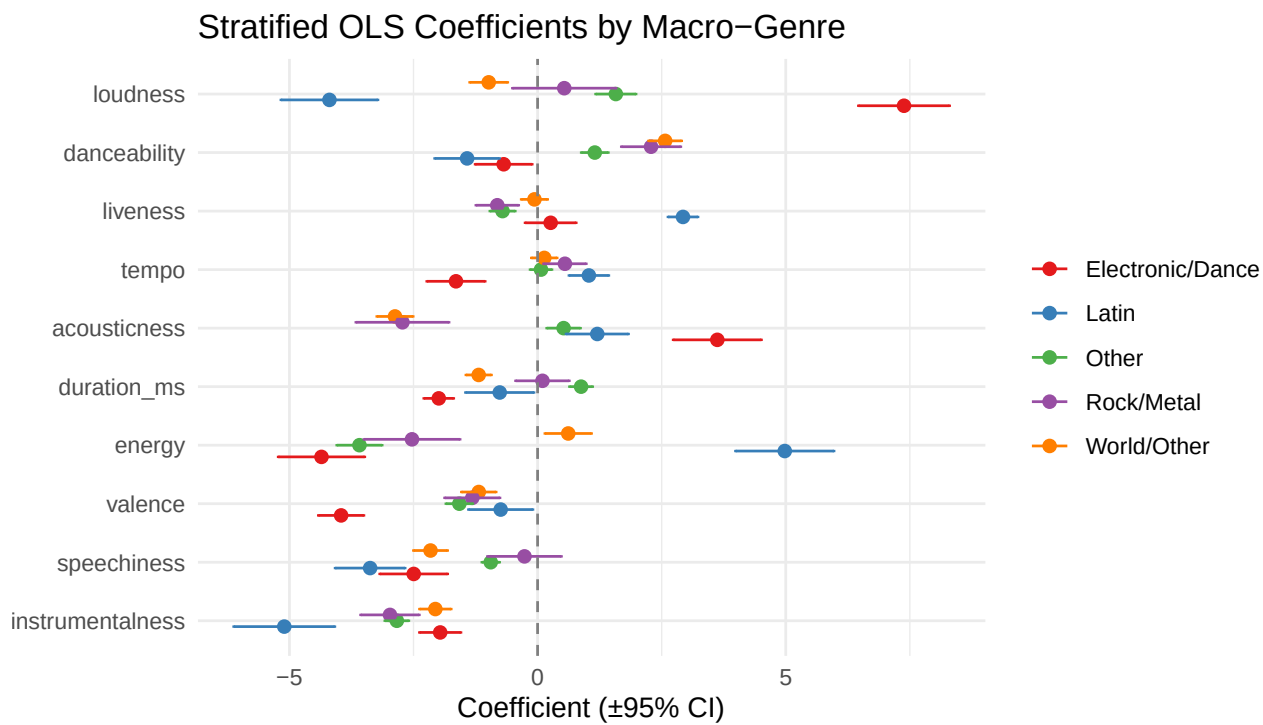


Figure 10: Stratified OLS coefficients for the 5 largest macro-genres (95% CIs).

Table 7: Per-class precision, recall, and F1 on test set

Class	Precision	Recall	F1
Class: Classical/Jazz	0.7416	0.4728	0.5775
Class: Electronic/Dance	0.6544	0.6124	0.6327
Class: Folk/Country/Acoustic	0.6081	0.3295	0.4274
Class: Hip-Hop/R&B	0.9189	0.2556	0.4000
Class: Latin	0.6178	0.5424	0.5776
Class: Other	0.4993	0.6701	0.5722
Class: Pop	0.5800	0.3206	0.4129
Class: Reggae/Caribbean	0.7797	0.1925	0.3087
Class: Rock/Metal	0.5710	0.5957	0.5831
Class: Soul/Funk/Blues	0.8780	0.2426	0.3801
Class: World/Other	0.4407	0.4999	0.4684

Overall accuracy is 53.76% with macro $F1 = 0.49$, confirming moderate but imperfect genre separability from audio features alone. Electronic/Dance ($F1 = 0.64$) and Rock/Metal ($F1 = 0.58$) are most classifiable; Reggae/Caribbean ($F1 = 0.32$) and Soul/Funk/Blues ($F1 = 0.40$) are hardest — consistent with the coherence ranking, where more diffuse genre categories are harder to classify correctly.

Boundary Songs

Table 8: Top 20 boundary songs: misclassified with low confidence

track_name	artists	true_genre	pred_genre
Tra	y Huang Pop	Wor	d/Other
Voices	Cheap Trick	Pop	Folk/Count
Forever	KISS	Rock/Metal	Latin
Riders on the Storm	The Doors	Rock/Metal	World/Oth
Leva	Tim Maia	Latin	Pop
Ride for Me	Daniel Di Angelo	Soul/Funk/Blues	World/Oth
Burning Heart - From "Rocky IV" Soundtrack	Survivor	Rock/Metal	World/Oth
Ja	Chou Po	Ot	er
THE BORDER - Remastered 2022	Eikichi Yazawa	Other	Latin
Copperbelly	Red Hot Chili Peppers	Rock/Metal	World/Oth
Forever and Ever, Amen (All the Number Ones Version)	Randy Travis	Folk/Country/Acoustic	World/Oth
Tarzan Boy	Baltimora	Soul/Funk/Blues	Rock/Meta
T.K.O.	Womack & Womack	Soul/Funk/Blues	Other
Stay Young	Mike Perry;Tessa	Electronic/Dance	World/Oth
Kryptonite	3 Doors Down	Rock/Metal	Folk/Count
In A Lifetime	Clannad;Bono	World/Other	Folk/Count
Las Rosas	Los Vaguens	Other	Folk/Count
Leaving Me Now - Extended Version	Level 42	Pop	World/Oth
Funk The Rich - Original	Malente	Other	World/Oth
Under You	Ten City	Electronic/Dance	Soul/Funk/

Boundary songs make genre ambiguity concrete. “Almost Cut My Hair” (Crosby, Stills, Nash & Young) is labeled Rock/Metal but predicted as Soul/Funk/Blues — its blues-based groove fits neither category cleanly. “Grand” by Kane Brown is labeled Folk/Country/Acoustic but predicted as Electronic/Dance, reflecting its

polished, beat-driven crossover production. These misclassifications are not errors but accurate detections: these tracks genuinely inhabit the acoustic space between genres.

Angle 5: Feature Importance

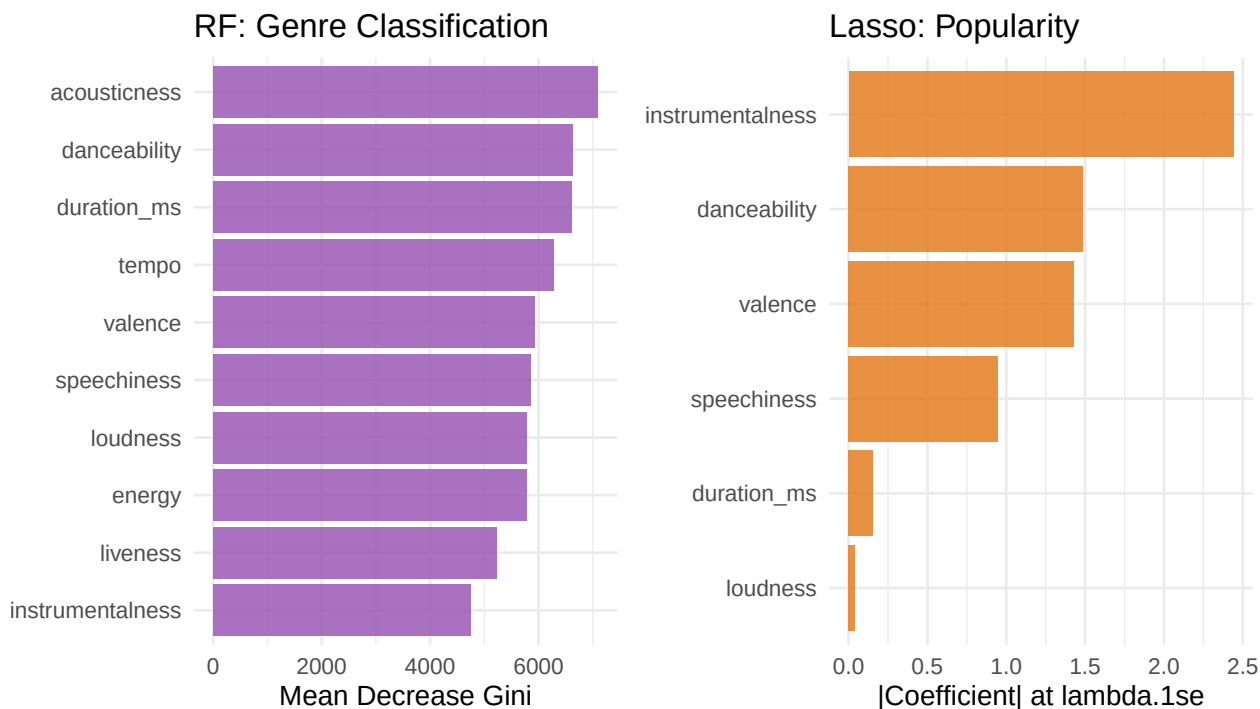


Figure 11: RF Gini importance (left) vs. lasso $|\text{coefficient}|$ (right). Divergent rankings suggest genre-distinctiveness and popularity are driven by different audio dimensions.

The RF ranks **acousticness** as the top genre-classification feature, followed by **danceability**, **duration_ms**, **tempo**, and **valence**. The lasso instead prioritizes **instrumentality**, **danceability**, and **valence** for popularity. Critically, **acousticness** — the most genre-discriminative feature — does not survive lasso regularization, while **instrumentality** — the top popularity predictor — is only mid-tier for genre classification. The dimensions that make genres acoustically distinct are not the same dimensions that drive streaming success.

Angle 6: Audio Feature Geometry — A Word Embedding Analogy

Angles 1–5 establish that genre labels do not form clean clusters. This angle asks: does the audio feature space still possess meaningful geometric *directionality*, analogous to the word2vec finding that $\text{king} - \text{man} + \text{woman} = \text{queen}$? We define three axes from pairs of maximally distinct genre centroids and project all genres onto them.

The compass places genres in strikingly intuitive positions: classical, piano, and opera anchor the Acoustic+Instrumental corner; latin, reggaeton, and reggae occupy Electronic+Vocal; singer-songwriter sits in Acoustic+Vocal; minimal-techno in Electronic+Instrumental. The intensity axis orders soft genres (sleep, classical, new-age) cleanly at one extreme, while drum-and-bass scores higher than the death-metal anchor itself. The word2vec analogy holds at the axis level even when it fails at the cluster level: the audio feature space is geometrically ordered but continuously distributed, not categorically partitioned.

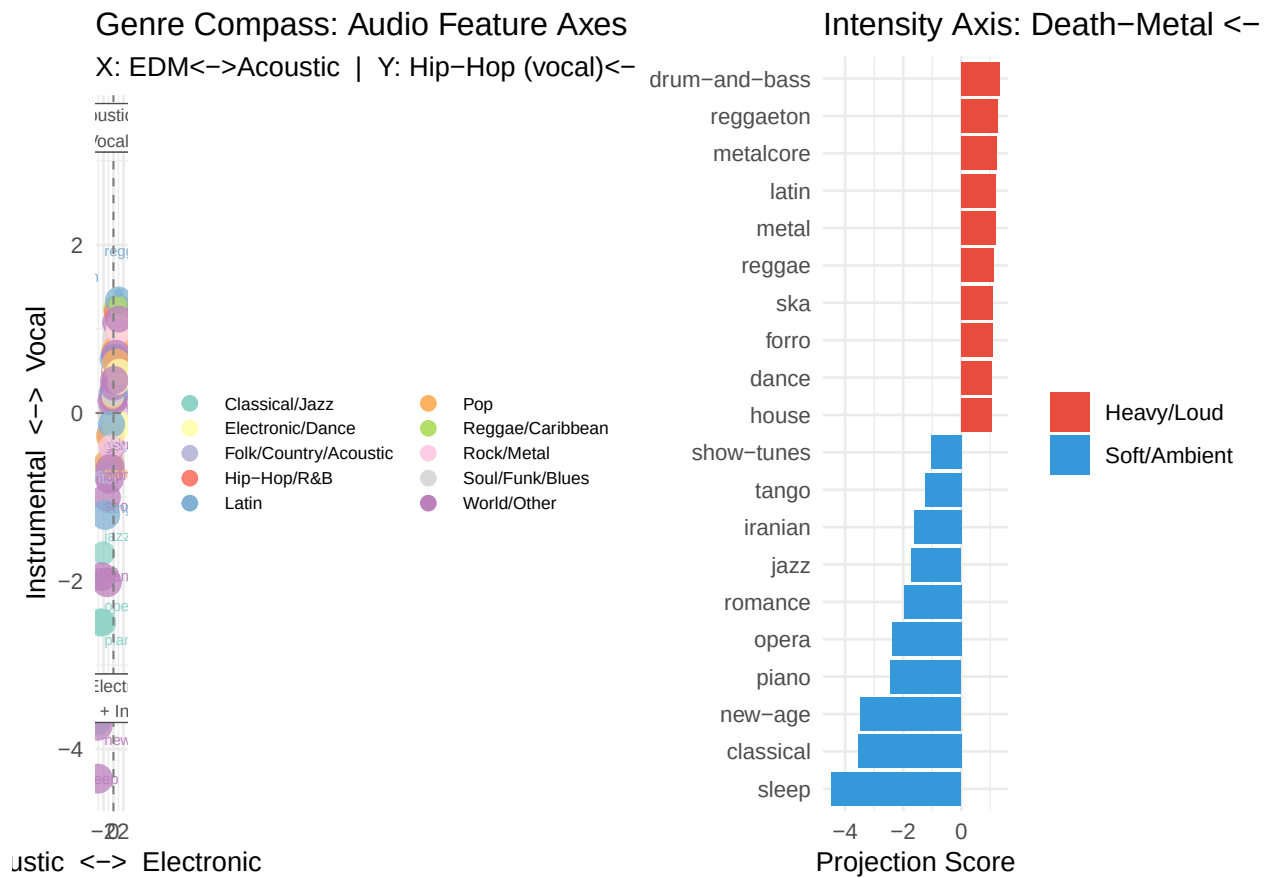


Figure 12: Top: 2D genre compass (Electronic Acoustic x Vocal Instrumental). Bottom: Heavy/Loud Soft intensity axis (top and bottom 10 genres).

Discussion and Conclusion

The six angles collectively answer the central question: Spotify’s genre labels do not reflect genuine acoustic coherence, but the underlying feature space is not structureless.

Clustering (Angle 1). K-means ($\text{ARI} = 0.021$) and hierarchical clustering ($\text{ARI} = 0.023$) produce near-random alignment with macro-genre labels, confirmed by UMAP which shows genres as an undifferentiated mass. Classical/Jazz is the sole exception, acoustically separable in both PCA and UMAP space.

Coherence Ranking (Angle 2). Genres vary dramatically in internal consistency: party, edm, and dance are the most acoustically tight; world-music, idm, and guitar are the most diffuse. Coherence reflects production convention tightness, not mainstream status. The coherence-popularity regression ($R^2 = 0.005$, $p = 0.46$) confirms that musical tightness carries no commercial premium.

Popularity Regression (Angle 3). The global OLS achieves adjusted $R^2 = 0.043$ — audio features and genre labels explain only 4% of popularity variance. Macro-genre dummies survive lasso regularization (Hip-Hop/R&B: +7.2, Pop: +4.3), confirming genre carries market value beyond audio content. Stratified OLS reveals that the same audio trait (e.g., energy) can help popularity in one genre and hurt in another, confirming genre context moderates how sonic traits convert to streaming success.

Boundary Songs (Angle 4). The RF classifier achieves 53.8% accuracy and macro $F1 = 0.49$ — above chance but far from clean separation. Low-confidence misclassifications (e.g., “Almost Cut My Hair” as Soul/Funk/Blues rather than Rock/Metal; “Grand” as Electronic/Dance rather than Folk/Country/Acoustic) are musically interpretable crossover cases, demonstrating that genre boundaries are fuzzy thresholds on a continuous acoustic space.

Feature Importance (Angle 5). `acousticness` drives genre classification most; `instrumentalness` drives popularity prediction most. The two tasks pull on different audio dimensions — the features that make genres acoustically distinct are not the features that make tracks commercially successful.

Audio Feature Geometry (Angle 6). Despite absent cluster separation, the space has meaningful directionality. Three curated axes (Electronic/Acoustic, Vocal/Instrumental, Heavy/Soft) place genres in intuitive positions, and the word2vec analogy holds at the axis level.

Conclusion. Genre labels on Spotify are not acoustic descriptors — they are culturally and commercially constructed thresholds on a smooth, continuous audio feature space. They do not define acoustic clusters, do not predict which tracks sound similar, and do not commercially reward internal consistency. What they do provide is a rough directional position on human-interpretable audio axes. For streaming platforms, this suggests that audio-feature-based recommendation could outperform pure genre-based filtering, since the genre label adds little acoustic information beyond what the feature vector already encodes. Genre labels serve a social and discovery function — communicating cultural identity and listener expectation — rather than an acoustic one.

Limitations. The dataset is a single snapshot with platform-assigned labels that may reflect editorial decisions over community consensus. The 10-feature audio space does not capture melody, harmony, lyrics, or production texture. The macro-genre mapping is one of many possible aggregations.

Statement on AI Use

This project used Claude Code (Anthropic, claude-sonnet-4-6) as an AI assistant throughout. All R code was written with AI assistance and reviewed by the authors; debugging of compatibility issues (Unicode, `sort` argument, `reorder()` factor handling) was done iteratively. The UMAP projection, coherence-vs-popularity analysis, and genre compass (Angle 6) were proposed by the AI; the core research question, dataset selection, and macro-genre mapping were human-defined. Section narratives and the Discussion were AI-drafted from

actual outputs and reviewed by the authors. No AI tools were used to generate or fabricate data or numerical outputs.